Enthusiast Course

PHASE : SPEED & ACCURACY TEST
TARGET : PRE-MEDICAL 2022

Test Type :**MINOR** Test Pattern :**NEET (UG)**

TEST DATE :

ANSWER KEY

Q.	1	2	3	4	5	6	7	8	9	10	11	12	13	14	15	16	17	18	19	20	21	22	23	24	25	26	27	28	29	30
A.	4	4	1	3	2	3	1	3	3	2	2	2	1	4	2	1	3	4	4	3	3	4	2	1	2	2	3	1	4	4

Q.	31	32	33	34	35	36	37	38	39	40	41	42	43
A.	3	1	4	2	2	1	3	3	3	2	3	4	3

HINT – SHEET

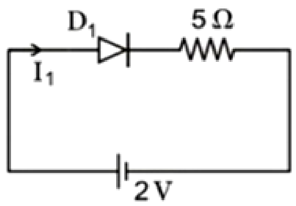
1. **Ans (4)**

In p- type semiconductor, holes are the majority and electrons are the minority charge carriers.

2. **Ans (4)**

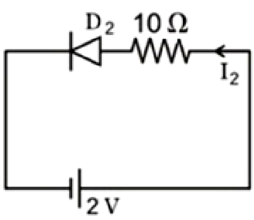
When +ve polarity is connected to A

Then D_1 is forward biased and D_2 is reversed biased i.e.



So I_1 current through $D_1 = \frac{2}{5} = 0.4$ A

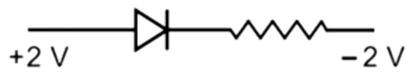
When -ve polarity is connected to B then D_2 is forward biased and D_1 is reversed biased then



$I_2 = \frac{2}{10} = 0.2$ A

3. **Ans (1)**

In case of forward biased circuit P-junction should be at higher potential therefore



4. **Ans (3)**

The Boolean expression for the given combination is output Y=(A+B).C Hence A=1, B=0, C=1 or other combination can be A=l, B =l, C=l

5. **Ans (2)**

The transistor conducts in active state in which input circuit is forward biased and output circuit is reversed biased

6. **Ans (3)**

$$E = \frac{12400}{24800} eV = 0.5eV$$

7. **Ans (1)**

$$R_L = \frac{V_{CC} - V_{CE}}{I_C} = 1k\Omega$$

$$\nu = \frac{I_C}{I_B} \Rightarrow I_B = 0.4\mu A$$

$$R_B = \frac{V_{CC} - V_{BE}}{I_B} = 185\ k\Omega$$

8. **Ans (3)**

Potential across Zener diode = 6V

$$I_L = \frac{6}{4} = 1.5mA$$

$$I_S = \frac{10}{2} = 5mA$$

$$I_Z = I_S - I_L = 3.5mA$$

10. **Ans (2)**

$$Y = A + \left(\bar{A} \bullet B\right) = \left(A + \bar{A}\right) \bullet (A + B)$$

$= 1 \cdot (A + B) = A + B$

The given circuit is equivalent to OR gate

11. **Ans (2)**

As the temperature is increased, more electrons are able to move from valance band to conduction band, hence the number of charge carriers increases but with increase in the temperature the frequency of collisions increases so the average drift speed decreases.

12. **Ans (2)**

Voltage gain-

$$\frac{V_0}{V_i} = \frac{R_0 \Delta I_c}{R_i \Delta I_B} = \frac{2000 \times 1.5 \times 10^{-3}}{150 \times 20 \times 10^{-6}}$$

$$= \frac{3}{3000 \times 10^{-6}} = 1000$$

13. **Ans (1)**

Forward bias causes a force on the electrons pushing them from the N side toward the P side. With forward bias, the depletion region is narrow enough that electrons can cross junction into p-type material.

14. **Ans (4)**

Given that, change in emitter current,

$\Delta I_E = 8mA$ and change in collector current,

$\Delta I_C = 7.9$ mA. We know that,

$$\alpha = \frac{\Delta I_C}{\Delta I_E}$$

$$\alpha = \frac{7.9}{8}$$

$\alpha \approx 0.99$

Also, we know that

$$\beta = \frac{\alpha}{1 - \alpha}$$

$$\beta = \frac{\frac{7.9}{8}}{1 - \frac{7.9}{8}} = \frac{7.9}{8 - 7.9} = \frac{7.9}{0.1} = 79$$

Hence the required answer is

$\alpha = 0.99$ and $\beta = 79$

15. **Ans (2)**

Input supply is e= 50sin (314t) Volt

So, frequency is $f = \frac{\omega}{2\pi} = \frac{314}{2\pi}$

$f = 50$ Hz

For full wave rectifier frequency of output wave is doubled the input frequency. So it will be 100Hz.

16. **Ans (1)**

For the voltage regulation Zener diode is connected in parallel to load resistance and is connected in reversed bias.

17. **Ans (3)**

Depletion layer in the p — n junction is a region which is depleted of charge carriers i.e., electrons and holes and it consists of positive and negative ions fixed in their position.

18. **Ans (4)**

Drif velocity $v_d = \frac{1}{nAe}$

$$\frac{(v_d)_{electron}}{(v_d)_{hole}} = \left(\frac{I_e}{I_h}\right)\left(\frac{n_h}{n_e}\right) = \frac{7}{4} \times \frac{5}{7} = \frac{5}{4}$$

19. **Ans (4)**

$n_i = 1.6 \times 10^{16}\ m^{-3}$

$n_e = 4.8 \times 10^{20}\ m^{-3}$

$n_h = ?$

The concentration of holes in the semiconductor is given by the expression

$n_i^2 = n_e \times n_h$

$(1.6 \times 10^{16})^2 = 4.8 \times 10^{20} \times n_h$

$n_h = \frac{2.56 \times 10^{32}}{4.8 \times 10^{20}} = 5.3 \times 10^{11}\ m^{-3}$

20. **Ans (3)**

$\overline{A + B} = Y_1$

$\overline{A + \bar{B}} = Y_2$

$Y = \overline{Y_1 + Y_2} = \overline{Y_1} \cdot \overline{Y_2}$

$Y = (A + B) . \left(A + \overline{B\backslash;}\right)$

$Y = \left(A + B\overline{B\backslash;}\right)$

$Y = A$

21. **Ans (3)**

The peak value of rectified output voltage

= pea k value of input voltage - barrier voltage

=2-0.7=1.3V

22. **Ans (4)**

Due to heating the number of electron-hole pair will increase, so overall resistance of diode will change. Due to which forward and reverse biasing both are changed. Hence, in a p-n junction diode, change in temperature due to heating affects the overall V-I characteristics of PN junction.

23. **Ans (2)**

$V_{BE} = V_{CC} - I_B R_B$

$= 5.5 - (10 \times 10^{-6}) \times (5 \times 10^5) = 0.5V$

24. **Ans (1)**

The operating point of transistor amplifier should be in the middle of its active region

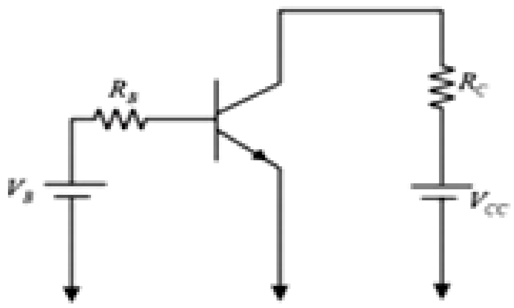
25. **Ans (2)**

for half – wave rectifier ripple frequency

= input frequency

= 50Hz

26. **Ans (2)**



Given , $\beta = 250$

$R_C = 1k\Omega = 1000\Omega$

$V_{CC} = 10V$

At saturation , $V_{CE} = 0$

$\Rightarrow V_{CC} - I_C R_C = 0$

$\Rightarrow i_C = \dfrac{10V}{1000\Omega} = 10mA$

Current gain factor,

$\beta = \dfrac{i_C}{i_B}$

$\Rightarrow i_B = \dfrac{i_C}{\beta} = \dfrac{10mA}{250} = 40\mu A$

27. **Ans (3)**

$\dfrac{I_C}{I_E} = \alpha$

$\beta = \dfrac{I_C}{I_B}$

28. **Ans (1)**

Given expression is

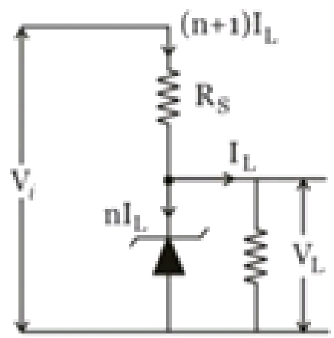
$\overline{(A + B)} \; . \; \overline{(A. B)}$

Substitute A=0 and B =0 in above expression

$\overline{(0 + 0)} \; . \; \overline{(0.0)} = \bar{0}\bar{0} = 1.1 = 1$

Hence (0,0) is the input of the given Boolean expression

29. **Ans (4)**



Across Zener diode V_L = constant

$V_i = V_L + (n+1) I_L R_s$

$V_i - V_L = (n+1) I_L R_s$

$R_s = \dfrac{V_i - V_L}{(n+1) I_L}$

30. **Ans (4)**

In a forward biased p-n junction, the applied potential is opposite to the junction barrier potential The consequence of this is the effective barrier potential reduces.

Hence, the graph 3 & 4 can be correct choice but as width of depletion layer is less in 4, hence 4 will be most appropriate choice.

31. **Ans (3)**

The energy band gap is maximum in insulators.

32. **Ans (1)**

For given electrons

Emitter current $I_e = \frac{n_e e}{t}$

Where n_e is the number of electrons enters into the emitter, e is the charge on electron and t is time.

$$I_e = \frac{10^{10} \times \left(1.6 \times 10^{-19}\right)}{10^{-6}}$$

$I_e = 1.6 \times 10^{-3} = 1.6\text{mA}$

2% electrons are lost in base region hence base current $I_b = 2\% \times 1.6 \times 10^{-3}$, therefore

$I_b = 0.032\text{mA}$

$I_c = I_e - I_b$

$I_c = 1.6 - 0.032$

$I_c = 1.568$ mA

And the current amplification factor

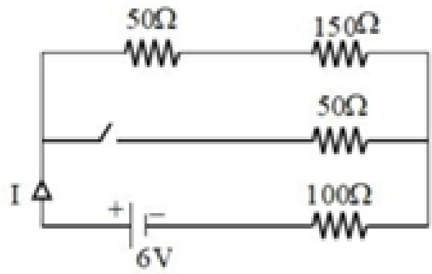
$$\beta = \frac{I_c}{I_b} = \frac{1.568}{0.032} = 49$$

33. **Ans (4)**

In a transistor base is thinnest and collector is largest therefore ℓ_2 is least and ℓ_3 is maximum.

34. **Ans (2)**

D_1 forward bias, D_2 reverse bias



$I = \frac{V_{net}}{R_{net}}$

$= 0.02\text{A}$

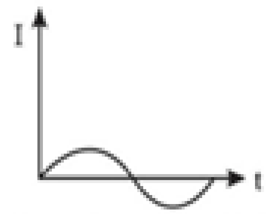
35. **Ans (2)**

$$R = \frac{\Delta V}{\Delta I} = \frac{2.3 - 0.3}{10 \times 10^{-3}}$$

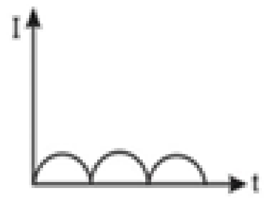
$$R = \frac{2}{10} \times 10^3$$

$R = 0.2 \times 10^3\,\Omega = 0.2\;k\Omega$

36. **Ans (1)**

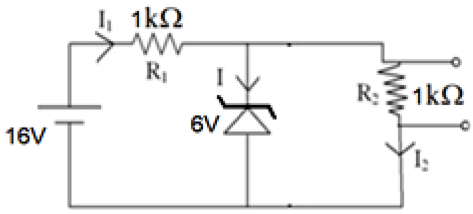


Alternating current (A.C)



D.C after rectifying action

37. **Ans (3)**



The potential difference across the Zener diode and the resistance R_2 is the same because they are in parallel connection.

$V_Z = V_2 = 6$ V

Let the potential difference across R_1 is V_1,

Then

$16 = V_1 + 6$

$V_1 = 10$V

$I_1R_1 = 10$

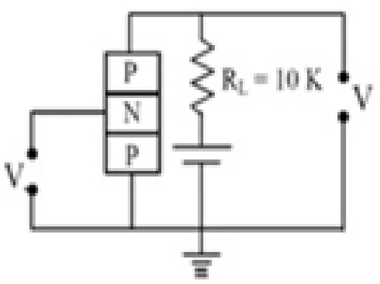
$I_1 = \frac{10}{1000} = 0.01$ A

$I_2R_2 = 6$V

$I_2 = \frac{6}{1000} = 0.006$ A

$I = 0.01 - 0.006 = 0.004$ A $= 4$ mA

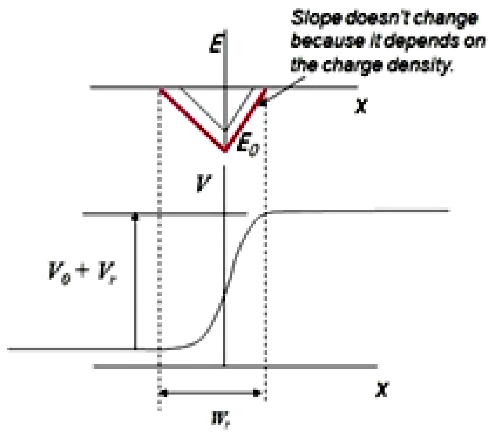
38. **Ans (3)**



The P-N junction which is forward biased is the emitter-base pair of the transistor. hence the common junction is collector, so it is a common collector circuit

39. **Ans (3)**

Electric field is maximum in the middle of the depletion layer of a reverse biased p-n junction.



40. **Ans (2)**

Forward bias resistance

$R_1 = \frac{\Delta V}{\Delta I} = \frac{0.1}{10 \times 10^{-3}} = 10\Omega$

Reverse bias resistance

$R_2 = \frac{10}{10^{-6}} = 10^7\,\Omega$

Ratio of resistances

$\frac{R_1}{R_2} = \frac{\text{Forward bias resistance}}{\text{Reverse bias resistance}} = 10^{-6}$

41. **Ans (3)**

$A = 1, B = 0, \bar{B} = 1$

$\therefore A + \bar{B} = 1 + 1 = 1 = A$

42. **Ans (4)**

The collector current

$I_C = \frac{2}{2 \times 10^3} = \frac{\Delta V}{R}$

$I_C = 1 \times 10?^{(-3)}$A

The base current

$I_B = \frac{I_C}{\beta} = \frac{10^{-3}}{200}$

$I_B = 5 \times 10^{-6}$A

$I_B = 5\mu$A

And

$V_B = I_BR_i = 5\mu A \times 1.5k\Omega = 0.0075$V

43. Ans (3)

The forbidden energy gap, $E_g = \frac{hc}{\lambda}$

λ = maximum wave length of radiation

$$\lambda = \frac{hc}{E_g} = \frac{12400}{0.72} = 17222\ \overset{o}{A}$$